# Huawei Access Node Routing Protocol Flap Resolution Procedure for KR-ACC-02

**Node:** KR-ACC-02 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4292

## Description

The KR-ACC-02, situated at KR Puram Access, functions as a critical access node within the network, facilitating the aggregation of subscriber traffic and its subsequent forwarding towards the core network. The observed Routing Protocol Flap, denoted by ALARM_4292 with a severity of Warning, is indicative of instability within the routing domain, potentially stemming from BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. This instability can arise due to a myriad of factors including, but not limited to, configuration discrepancies, hardware malfunctions, or issues with the underlying network topology. The Huawei equipment, with its proprietary CLI dialect, necessitates a nuanced approach to diagnosis and resolution, leveraging commands and techniques specific to the vendor's implementation of routing protocols. The resolution of such issues is paramount to ensure the uninterrupted flow of traffic and to maintain the overall health and reliability of the network.

## Pre-requisite Checks

- Verify the physical integrity of all connections and cables at KR Puram Access
- Confirm the operational status of redundant links to ensure no single point of failure
- Dispatch remote hands to the site for hands-on troubleshooting and potential hardware replacement
- Ensure all necessary tools and spares are available on site

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfiguration of BGP attributes leading to route churn | High |
| Hardware failure affecting OSPF adjacency establishment | Medium |
| ISIS LSP flooding due to network topology changes | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display ip bgp peer, Expected Output: List of BGP peers with their current status, Description: This command isolates issues in the BGP control plane by displaying the status of all BGP peers, helping to identify any peers that are not in the expected state, such as 'Established'.

Command: display ospf peer, Expected Output: List of OSPF neighbors with their current state, Description: By examining the state of OSPF neighbors, this command aids in diagnosing adjacency issues, such as neighbors stuck in the 'Init' or '2-Way' state, indicating a problem with OSPF hello packets or the network topology.
```

```
Command: display isis lsdb, Expected Output: ISIS link-state database contents, Description: This command provides insight into the ISIS link-state database, facilitating the identification of potential issues such as LSP flooding, which could be indicative of network instability or misconfiguration.
```

## Resolution Steps

Step 1: Step: Adjust BGP Attributes, Command: bgp peer ebgp-max-retain , Description: Adjusting BGP attributes such as the eBGP maximum retain value can help stabilize BGP routes and prevent churn. This step involves modifying the BGP configuration to better match the network's specific requirements and topology.

Step 2: Step: Reset OSPF Adjacency, Command: reset ospf process, Description: In cases where OSPF adjacency loss is identified, resetting the OSPF process can help reestablish adjacencies. This step should be taken with caution, as it may temporarily disrupt network stability.

Step 3: Step: Optimize ISIS Configuration, Command: isis enable level-1, Description: Optimizing the ISIS configuration, such as enabling level-1 routing for the appropriate process ID, can help mitigate issues related to LSP flooding and improve network convergence times.

## Post-Check Verification

```
Command: display ip routing-table, Expected Output: Updated IP routing table reflecting changes, Description: After implementing resolution steps, verifying the IP routing table ensures that the changes have taken effect and the routing protocol flaps have been addressed.

Command: display bgp routing-table, Expected Output: BGP routing table with stabilized routes, Description: This command confirms that BGP routes have stabilized and are no longer experiencing churn, indicating a successful resolution of the identified issues.
```

## Roll-back Procedures

- Step: Revert BGP Configuration, Command: undo bgp peer ebgp-max-retain, Description: If the adjustment of BGP attributes does not yield the desired outcome, reverting these changes is crucial to prevent further instability. This step involves removing the recently added configuration to return to the previous state.

- Step: Restore Original OSPF Configuration, Command: ospf abr-priority , Description: If resetting the OSPF process or adjusting its configuration does not resolve the issue, or if it introduces new problems, restoring the original OSPF configuration settings is necessary to maintain network stability.